

Element K: Reflection on the Design Project

Engineering Design Process Summary:

During the time that we have been creating this product we have gone through several processes in order to reach the point we are at now. At the start of our project we decided to focus on the problem of kidnapping. Our idea was to create a device that could help reduce the cases of kidnapping by creating products with a tracker installed on them. Before we got settled on one of the designs or ideas for this tracker, we decided to brainstorm individually and make up our own designs and ideas. From there we would draw these designs and give measurements and specifics of the design. Once we were all done brainstorming individually, we came together and chose on a design that focuses on our main priority for this project. This priority was making sure that the device was able to provide a live location of the person wearing this device.

As part of our design process, we decided to contact a stakeholder to help us with the design of the device as well. At first, we chose a design that used a micro bit as a tracker. After talking to the stakeholder he advised that it was better to use an actual tracker instead of the micro bits. Of course, this affected the materials we used in order to make the prototype of the device. The first prototype we created involved a belt, tape, and a tracker. First, we cut some duct tape into a small square portion. Second, we made a pocket made out of duct tape around the belt, making sure the pocket was big enough to fit the tracker in it. Third, you place the tracker in the pocket and attach the pocket to the belt. The second prototype involved a watch, duct tape, and a tracker. The steps for creating this prototype are the same as the belt prototype mentioned before.

Once we finished without prototypes, we went on to testing these prototypes, wanting to make sure they met our main focus. For testing, we had to make sure the tracker on the prototypes were turned on and ensure that the tracker actually recorded the live location of the person wearing it. We also had to make sure that the prototype was able to withstand damage done to it, such as dropping it from a certain height. Once we finished testing the prototype, we were satisfied with the results that it had shown us.